Lecture 10: Evolution, Emotion, and Reason: Evolution and Rationality

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-10 (CC BY-NC-SA 3.0).

Before you start

This lecture introduces evolutionary psychology: the idea that natural selection shaped the mind, as it shaped the body. Professor Bloom explains common mistakes and shows how evolution gives new insights. Then he begins the topic of rationality: mental shortcuts that can mislead us.

Learning goals

After this lecture you should be able to:

- Explain Paley's argument from design and how Darwin answered it.
- Explain the difference between ultimate and proximate causation.
- Tell the difference between adaptations and accidents, with examples.
- Give two examples of how asking "What is it for?" leads to new ideas.
- Describe four heuristics or biases, with examples.

Key ideas

1. The Modern Biological Account of the Origin of Psychological Phenomena

- The course's first big idea was that the mind is the product of the brain. The second is what Daniel Dennett called "Darwin's dangerous idea": a purely physical account of where living things, including minds, come from.
- William Paley's **argument from design**: if you find a watch, you assume someone made it. The eye is like a camera but far more complex, so it must have a designer, God. This view is called creationism.
- Problems: it pushes the question back (where did the designer come from?), and there was evidence for evolution: fossils, vestigial features (leftovers from earlier forms, like the tailbone), similar brains in different animals, and poor design (the blind spot in the eye).
- Richard Dawkins wrote that before Darwin, the argument from design was a very good argument. Darwin showed how complex structures come from a purely physical process, **natural selection**: variation, differences in survival and reproduction that are passed on, and the result, adaptations such as camouflage.
- Put this together with the idea that the mind is physical, and our mental abilities must also come from natural selection, for survival and reproduction.

2. Avoiding Misconceptions When Applying Evolutionary Theory to Psychology

- **Mistake 1:** thinking that people consciously want to spread their genes. We must separate **ultimate causation** (why a trait exists, over millions of years) from **proximate causation** (why you do something now). William James: a man "eats because the food tastes good and makes him want more." We protect our children because animals that didn't protect their offspring did not last, but nobody says, "I love my children because they contain fifty percent of my genes."
- **Mistake 2:** thinking everything is adaptive. Some things are byproducts or accidents, like back pain or hiccups. We evolved a taste for sweet things like fruit; liking chocolate is an accident of the modern world.
- Clear adaptations: color vision, fear of snakes. Hard cases: music (Steven Pinker called it "auditory cheesecake," an accident; others disagree), visual art and fiction. The real work is figuring out which is which.

3. Claims against the Evolutionary Psychology

Some people reject evolutionary psychology because they are dualists (they believe the mind is not physical): Pope John Paul II accepted evolution of the body, but said the soul is created by God. Others say humans have no instincts apart from a few infant reactions (Ashley Montagu, 1973), or that the most interesting parts of human life are cultural (Louis Menand). A third view: we can just study people, so evolution adds nothing.

4. Ways in Which Evolution Helps Describe the Mind

- It shows what can be innate. Old problems (talking, mate selection, childcare) could lead to special adaptations. Recent ones (writing, driving, chess) could not, so there cannot be an evolved "chess part" of the brain.
- It predicts group differences: children should differ from adults. It gives no adaptive reason to expect profound racial or ethnic differences in mental abilities. It can also help us judge claims about sex differences.
- Asking "What is it for?" helps. Margie Profet proposed that pregnancy sickness protects the developing baby from harmful foods. Her theory made predictions: its timing should match the baby's most vulnerable period, and more sickness should mean more protection. Women don't develop an aversion to alcohol because alcohol was not around during our evolutionary history.
- Salovey's "big three" (proximity, similarity, familiarity) fail for siblings. Evolution predicts **incest avoidance**, because mating with close relatives produces unhealthy offspring. We feel "Eew" (proximate); we don't calculate (ultimate). Kibbutz studies suggest we are not attracted to people we grew up with. A paper in *Nature*, published days before the lecture, found an even bigger cue: seeing your mother take care of that person.

5. Heuristics: Framing Effects, Base Rates, Availability Bias and Confirmation Bias

- Danny Kahneman won a Nobel Prize for work with Amos Tversky showing that we use **heuristics**, rough mental shortcuts that served us during evolution but can mislead us.
- **Framing effects:** a disease will kill 600 people. Most people prefer a program that saves 200 for sure over a gamble (a one-third chance that everybody is saved). Describe the same choice in terms

of deaths, and choices flip. People hate a certain loss (loss aversion). Laurie Santos found that capuchin monkeys also show framing effects. In the **endowment effect**, people offer about $2 for a chocolate bar but want about $2.50 to sell one they own.

- **Base rates:** people decide "John" is an engineer from his description, whether the group has 70 or 30 engineers. In a disease example (1 in 1,000 people have it; the test never misses, but says "yes" for 5% of healthy people, a false positive), medical students guessed a 50-95% chance of disease after a positive test. The answer is about 2%.
- **Availability bias:** we judge frequency by how easily examples come to mind. People estimate more English words ending in "ing" than in "ng," although every "ing" word also ends in "ng." Shark attacks are always in the news, so people overestimate them (deaths are about one in 500 million), but food poisoning is far more dangerous. Americans guess that about 20% of the US is Jewish; the real figure is about 2%. One reason: Jewish people are prominent in visible fields like entertainment.
- **Confirmation bias:** we look for evidence that confirms our hypothesis. In a custody case, people were more likely both to award custody to parent B and to deny custody to parent B. In the Wason selection task (cards D, G, 3, 8), people find the logic hard. Leda Cosmides showed it becomes easy as a social rule: a bartender checking that nobody under twenty-one drinks beer.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
argument from design	complex things must have an intelligent designer	Paley's watch and the eye
vestigial	left over from an earlier form; no longer useful	the tailbone, goose bumps
natural selection	variation + differences in survival and reproduction that are passed on	camouflage, the eye
adaptation	a trait shaped by natural selection because it helped survival or reproduction	color vision, fear of snakes
byproduct / accident	a trait with no adaptive purpose	back pain, liking chocolate
ultimate causation	why a trait exists, over evolutionary history	protecting offspring who share our genes
proximate causation	the immediate reason you do something now	eating because food tastes good
heuristic	a rough mental shortcut	judging frequency by what comes to mind
framing effect	a choice changes when the same options are described differently	"80% fat free" vs. "20% fat"
endowment effect	we value something more once we own it	$2 to buy vs. $2.50 to sell

Term	Meaning in simple English	Example from the lecture
base rate	how common something is in the whole group	1 in 1,000 people have the disease
availability bias	judging frequency by how easily examples come to mind	sharks vs. food poisoning
confirmation bias	looking for evidence that supports what we already think	the custody case

Check your understanding

1. What was Paley's argument from design?
2. Name two kinds of evidence for evolution from the lecture.
3. Explain ultimate and proximate causation with the example of eating.
4. Is liking chocolate an adaptation? Explain.
5. Why can there be no evolved part of the brain for playing chess?
6. How does Margie Profet explain pregnancy sickness? Why don't women develop an aversion to alcohol?
7. In the disease test example, why is the chance of having the disease after a positive test only about 2%?
8. Which bias explains why people fear sharks more than food poisoning?

Answer guide

1. A watch must have a maker, and the eye is even more complex. So, Paley argued, living things must have an intelligent designer, God.
2. Any two: fossils, vestigial features, similar brains in different animals, poor design (the blind spot).
3. Ultimate: over evolution, eating kept our ancestors alive so they could reproduce. Proximate: you eat because you are hungry and food tastes good.
4. No. We evolved a taste for sweet things like fruit. Chocolate is a modern invention, so liking it is an accident.
5. Chess is a recent invention. Evolution had no time to build a brain part for it.
6. It protects the baby from foods that could harm it. Alcohol did not exist during our evolutionary history, so no protection against it evolved.
7. Of 1,000 people, 1 has the disease and tests positive. Of the other 999, about 50 (5%) test positive by mistake. So only 1 of about 51 positives is real.
8. The availability bias. Shark attacks are always in the news, so examples come to mind easily.

English practice

Useful phrases from the lecture

- "They're not alternative. Rather, they're different levels of explanation." - shows that two ideas do not compete.
- "Those are easy questions. Here are some hard questions." - moves from simple to difficult cases.
- "If she just ended there it's a story. How do we know it's true?" - asks for evidence.
- "The answer is an easy one." - introduces a simple explanation.
- "In general, ..." - states a general rule after examples.

Speaking task (with your AI tutor)

- Explain in 4-6 sentences why siblings are not attracted to each other. Use: *incest avoidance*, *ultimate*, *proximate*.

Writing task

- Choose one heuristic or bias from the lecture. In one paragraph (80-120 words), explain it, give the lecture's example, and say how it could cause a bad everyday decision.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 10. Ask me questions about adaptations and accidents, one at a time."
- "Quiz me on the key vocabulary of Lecture 10."

- "I will explain ultimate and proximate causation in my own words. Check my understanding and my English."

Readings from the original course

- Gray, Peter. Psychology (5th edition), pp. 61-68, 345-351
- Pinker, Steven. "How the Mind Works." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 39-44

My notes